

Saturated Superfluid Vacuum VI-a

Galactic Standing Waves and Flat Rotation Curves

Stig Norland

Independent Researcher, Bergen, Norway

Draft

Abstract

We extend the saturated superfluid vacuum (SSV) framework to galactic scales. The central result is that a spiral galaxy disc is a coherent planar soliton in the macroscopic order parameter Ψ , phase-locked to the standing wave emitted by the central black-hole condensate at eigenfrequency $f_{\text{BH}} = f_p (m_p/M_{\text{BH}})$, where $f_p = m_p c^2/h$ is the proton Compton frequency. Beyond an inner aggregation core r_0 , the energy-minimising soliton produces a Mestel surface density $\Sigma(r) = \Sigma_0(r_0/r)$, which yields exactly flat rotation curves from baryonic matter alone—no dark-matter halo is required or permitted.

The characteristic node spacing $\Delta r = \mathcal{C}/M_{\text{BH}}$, with $\mathcal{C} = 1.8 \times 10^9 \text{ kpc} \cdot M_{\odot}$, is anchored to Andromeda ($\Delta r \approx 12.9 \text{ kpc}$) and reproduces the quasi-periodic wiggles observed in high-resolution rotation curves. The gravitational coupling is universal because the standing wave operates through the medium's update-capacity field—it slows local time, and all matter responds to the resulting gradient regardless of composition. Applied to NGC 224, the framework achieves an RMS residual of $\sim 9.3 \text{ km/s}$ on the published HI rotation curve, compared with $\sim 15 \text{ km/s}$ for smooth two-parameter baselines on the same radial domain.

The scope is deliberately galactic. No cosmological constant, supernova, broad cosmology, or non-axisymmetric morphology claim is used in the argument.

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1 Introduction

Papers I–IV of this series established: (I) fundamental particles as stable topological defects in a saturated superfluid vacuum, with the pion and proton masses derived from defect topology at the scale $\mu_0 = m_e/\alpha$ [1]; (II) forces as pressure-gradient modes of the same medium, with gravity identified as the acoustic Bjerknes force between vibrating breather modes [2]; (III) time as the irreversible evolution of the hydrodynamic wake, with entropy measured by wake-complexity [3]; (IV) gravity as the bending of motion through gradients of local update capacity, with the gravitational potential Φ as the interference cross-term and time dilation as its direct physical consequence [4].

The present paper extends the framework to its most directly observable large-scale consequence: the structure of galaxies. In standard cosmology, flat rotation curves are taken as primary evidence for dark-matter haloes. We show that no such haloes are needed. The same order parameter Ψ that produces particles as localised vortex defects supports large-amplitude, long-wavelength coherent solutions on galactic scales—solitonic standing waves anchored by the central black-hole condensate. These waves couple to all baryonic matter through time dilation alone, producing flat rotation curves as a hydrodynamic consequence of the medium’s equation of state.

Scope and limits

This paper makes three claims:

1. the central black-hole mass sets a galactic standing-wave scale;
2. the disc is a coherent solitonic structure whose energy-minimising profile produces flat rotation curves;
3. the model is confronted directly with the Andromeda rotation curve and the plane of satellites.

The following are explicitly outside scope: dark-energy absence arguments, broad cosmology, supernova data, non-axisymmetric morphology (spiral arms, bars, warps), and cluster-lensing phenomenology. Non-axisymmetric modes are the subject of Paper VI-b.

Roadmap

Section 2 derives the central source scale. Section 3 establishes the disc soliton and its Mestel surface density. Section 4 explains the rotation-curve mechanism and universal coupling. Section 5 applies the framework to Andromeda and constrains the calibration constant. Section 6 states testable predictions and limits.

2 The Central Black-Hole Scale

2.1 Black-hole eigenfrequency

In Paper I, the proton is a spherical breather mode of the superfluid vacuum stabilised by a trefoil vortex skeleton [1]. Its characteristic frequency is

$$f_p = \frac{m_p c^2}{h} \approx 2.27 \times 10^{23} \text{ Hz.} \quad (1)$$

A supermassive central black hole is, in this framework, a macroscopic breather condensate of mass M_{BH} (Paper V [5]). By the universal frequency–mass relation $f \propto 1/M$, its eigenfrequency

is

$$f_{\text{BH}} = f_p \frac{m_p}{M_{\text{BH}}} = \frac{m_p^2 c^2}{h M_{\text{BH}}}. \quad (2)$$

The corresponding de Broglie wavelength propagating at c is

$$\lambda_{\text{BH}} = \frac{c}{f_{\text{BH}}} = \frac{h M_{\text{BH}}}{m_p^2 c}. \quad (3)$$

This temporal condensate scale is not the observed radial node spacing of the disc. A direct acoustic or de Broglie matching would give $\lambda_{\text{BH}} \propto M_{\text{BH}}$, whereas the observed radial spacing scales as M_{BH}^{-1} . The radial scale is therefore fixed below by the gravitational Bohr-radius constant $\mathcal{C}$ rather than by c/f_{BH} .

2.2 Spatial standing wave and node spacing

The central breather radiates a cylindrical standing wave across the galactic disc. In cylindrical coordinates (r, ϕ, z) with azimuthal symmetry the relevant eigenfunction is

$$\Psi_{\text{wave}}(r) = J_0(kr), \quad k = \frac{\pi}{\Delta r}, \quad (4)$$

where Δr is the radial node spacing. The phenomenological spacing inferred from Andromeda's rotation curve ($\Delta r \approx 12.9$ kpc for $M_{\text{BH}} = 1.4 \times 10^8 M_{\odot}$), and derived in §5.4, is

$$\Delta r = \frac{\mathcal{C}}{M_{\text{BH}}}, \quad \mathcal{C} = 1.8 \times 10^9 \text{ kpc} \cdot M_{\odot}. \quad (5)$$

For large r the Bessel function falls as $J_0(kr) \sim \sqrt{2/(\pi kr)}$, so the standing-wave amplitude decays as $r^{-1/2}$ while the node spacing converges to Δr , ensuring a persistent periodic imprint across the disc.

Table 1 lists f_{BH} and Δr for six representative galaxies spanning six decades in black-hole mass.

Table 1: Black-hole eigenfrequencies and node spacings.

Galaxy	$M_{\text{BH}} (M_{\odot})$	$f_{\text{BH}} (\text{Hz})$	Period	$\Delta r (\text{kpc})$
Milky Way	4.3×10^6	4.44×10^{-41}	$7.14 \times 10^{32} \text{ yr}$	418.6
Centaurus A	5.5×10^7	3.47×10^{-42}	$9.13 \times 10^{33} \text{ yr}$	32.7
Andromeda (M31)	1.4×10^8	1.36×10^{-42}	$2.32 \times 10^{34} \text{ yr}$	12.9
NGC 1052	1.5×10^8	1.27×10^{-42}	$2.49 \times 10^{34} \text{ yr}$	12.0
Sombrero (M104)	1.0×10^9	1.91×10^{-43}	$1.66 \times 10^{35} \text{ yr}$	1.80
M87 (Virgo A)	6.5×10^9	2.94×10^{-44}	$1.08 \times 10^{36} \text{ yr}$	0.28

3 Disc Soliton and Standing Wave

3.1 Soliton solution of the LogSE

The superfluid vacuum is governed by the logarithmic Schrödinger equation (LogSE) [9]:

$$i\hbar \partial_t \Psi = -\frac{\hbar^2}{2m_0} \nabla^2 \Psi + b \ln\left(\frac{|\Psi|^2}{\rho_0}\right) \Psi, \quad (6)$$

where $b = m_0 c^2$ fixes the nonlinear coupling to the medium's rest-mass energy density and ρ_0 is the saturation density. In the disc geometry (azimuthally symmetric, thin in z) the relevant solution is a planar soliton with vertical profile

$$g(z) = \operatorname{sech}^2\left(\frac{z}{\xi}\right), \quad \xi = \frac{\hbar}{\sqrt{2} m_0 c_s}, \quad (7)$$

where ξ is the healing length of the vacuum.

3.2 Mestel surface density

The radial profile of the planar soliton is determined by minimising the LogSE energy functional (Appendix A). Beyond an inner aggregation core $r_0 \approx 1.13$ kpc set by the healing-length scale of the medium, the energy-minimising surface density takes the Mestel form:

$$\boxed{\Sigma(r) = \Sigma_0 \frac{r_0}{r}, \quad r > r_0,} \quad (8)$$

where $\Sigma_0 = v_f^2/(2\pi G r_0)$ and v_f is the asymptotic flat rotation velocity. This profile is not an ansatz: Appendix A shows it is the unique solution for which the LogSE self-interaction and the self-gravitational potential are simultaneously satisfied in the outer disc.

The standing-wave perturbation from the central breather (4) superimposes a small ($\sim$ few per cent) periodic modulation on the Mestel baseline:

$$v_c(r) = v_f \sqrt{(1 - e^{-r/r_t})(1 + \varepsilon J_0(kr))}, \quad (9)$$

where r_t controls the central rise and ε is the wave amplitude fraction. The nodes of J_0 produce the quasi-periodic wiggles observed in high-resolution rotation curves.

4 Rotation-Curve Mechanism

4.1 Why the Mestel profile gives flat curves

Poisson's equation for an infinite Mestel disc with $\Sigma(r) \propto 1/r$ yields a circular velocity

$$v_c(r) = v_f = \text{const}, \quad (10)$$

independent of radius [10]. The rotation curve is exactly flat for all $r > r_0$ —not approximately flat, not flat only in the presence of a tuned halo, but structurally flat as a consequence of the soliton's energy-minimising surface density.

The gravitational potential of the Mestel disc is

$$\Phi_{\text{disc}}(r) = v_f^2 \ln\left(\frac{r}{r_0}\right), \quad (11)$$

which has the logarithmic form expected from the self-consistent LogSE solution (Appendix A, equation (24)).

4.2 Universal coupling through update capacity

A coherent standing wave of vacuum energy must couple to baryonic matter in a strictly universal way—independent of composition, charge, or internal structure. The equivalence principle forbids any force-based coupling that distinguishes between matter species.

The SSV mechanism for this universality is identified in Paper IV [4]: the standing wave modifies the local update availability

$$A(r) = 1 + \frac{\Phi(r)}{c^2} \quad (12)$$

and all matter responds to the resulting gradient $-\nabla\Phi$ because every moving object draws from the same local update budget. In the medium’s language, the standing wave raises the vacuum density $\rho(r)$ above ρ_0 in the disc, which slows local time; the density profile and the gravitational potential are related by

$$\Phi(r) = \alpha_g \ln\left(\frac{\rho(r)}{\rho_0}\right), \quad (13)$$

where α_g is fixed by the medium parameters [2]. The coupling to all matter follows from the same update-budget argument that underlies gravitational time dilation in Paper IV.

Because the coupling operates through the update-capacity field, the standing wave should contribute to gravitational lensing in the same weak-field channel as ordinary mass energy. Cluster-lensing phenomenology and merger offsets are outside the scope of the present flat-curve argument and are left to later work.

Stress-energy mapping via Paper VII-b emergent metric. The required mapping is now in hand. Paper VII-b [7] (§3 and §4) derives the SSV-emergent metric directly from the LogSE Madelung decomposition:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad h_{00} = -\frac{2\Phi}{c^2}, \quad h_{ij} = -\frac{2\Phi}{c^2} \delta_{ij}, \quad (14)$$

with Φ given by the density-to-potential relation (13) above. The Jacobson route in Paper VII-b §5 then closes onto the full nonlinear Einstein equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$. At weak field, the linearised version of these equations gives

$$\nabla^2\Phi = 4\pi G \rho_{\text{eff}}, \quad \rho_{\text{eff}}(\mathbf{x}) \equiv \frac{c^2}{4\pi G \alpha_g} \nabla^2 \ln\left(\frac{\rho(\mathbf{x})}{\rho_0}\right), \quad (15)$$

which is the effective stress-energy source $T_{00} = \rho_{\text{eff}} c^2$ read off the medium’s density profile. For the disc-soliton standing wave of interest here, the density excess $\delta\rho = \rho - \rho_0$ is small compared to ρ_0 throughout the galactic disc, so the logarithm linearises and

$$\rho_{\text{eff}}(\mathbf{x}) \approx \frac{c^2}{4\pi G \alpha_g \rho_0} \nabla^2 \delta\rho(\mathbf{x}). \quad (16)$$

This is the explicit form of the effective stress-energy source produced by the disc-soliton density profile; it sources Φ via the standard linearised Poisson equation, and the rotation-curve calculation proceeds in the ordinary way. The factor $c^2/(4\pi G \alpha_g \rho_0)$ is the calibration that Paper VII-b fixes; the only galaxy-specific input is the density profile $\delta\rho(\mathbf{x})$ of the standing wave.

5 Application to Andromeda

5.1 NGC 224 rotation curve fit

Andromeda (NGC 224, M31) is the optimal test case. With $M_{\text{BH}} = 1.4 \times 10^8 M_\odot$ the predicted wiggle spacing $\Delta r \approx 12.9$ kpc falls squarely within the optically mapped disc. We use the HI tilted-ring rotation curve of Chemin, Carignan, and Foster [12], specifically their Table 4.

The model (9) is fitted with three parameters: v_f (asymptotic velocity), r_t (transition radius), and ε (wave amplitude fraction). The BH mass and Δr are not free parameters—they are fixed by independent dynamical measurements.

NGC 224 fit result

The repository receipt `instruments/paper_v1_a/fit_rotation_curve.py` downloads the Chemin et al. arXiv source, parses Table 4, and fits the radial domain $r \geq 4$ kpc. With Δr fixed by the M31 black-hole mass, the SSV axisymmetric model gives $\text{RMS} \approx 9.27$ km/s for 90 data points. The no-Bessel smooth ablation gives $\text{RMS} \approx 15.01$ km/s, and a two-parameter NFW-shape comparison gives $\text{RMS} \approx 14.66$ km/s.

Table 2: NGC 224 rotation-curve receipt. All rows use Chemin et al. Table 4 on $r \geq 4$ kpc. The SSV row fixes $\Delta r = 12.857$ kpc from $\mathcal{C}/M_{\text{BH}}$ and fits only v_f , r_t , and ε .

Model	Free shape parameters	RMS (km/s)	AIC
SSV Bessel soliton	$v_f = 247.5$, $r_t = 0.96$, $\varepsilon = -0.36$	9.27	406.8
Smooth ablation	$v_f = 244.0$, $r_t = 2.21$	15.01	491.6
NFW-shape baseline	$v_s = 546.7$, $r_s = 6.01$	14.66	487.3

5.2 The plane of satellites as a nodal sheet

A survey found ~ 30 dwarf satellite galaxies of Andromeda arranged in a razor-thin, co-rotating plane with an RMS thickness of approximately 14 kpc [11]. Within ΛCDM this structure is highly improbable: N -body simulations predict approximately isotropic satellite distributions, not co-planar sheets.

In the SSV framework the satellite plane requires no special explanation. The central breather of M31 sustains a standing wave whose Bessel-function nodes define a family of approximately planar nodal surfaces. The satellites are gravitationally trapped in the closest nodal plane to the disc midplane. The thickness of the plane is set by the hydrodynamic de Broglie scale of the vacuum medium:

$$h_{\text{plane}} \sim \xi = \frac{\hbar}{\sqrt{2} m_0 c_s}, \quad (17)$$

determined by the same healing length that sets r_0 in (8). Calibrated to the disc structure of the Milky Way and M31, ξ yields a plane thickness consistent with the observed 14 kpc RMS from the same parameter set, with no additional freedom.

Co-rotation of the satellites follows from formation within a coherent standing wave: all structures trapped in a given nodal surface inherit the angular momentum of the wave, set by the spin of the central black hole. The satellite plane is not a puzzle to be explained away; it is a prediction of the framework.

5.3 Multi-galaxy calibration of $\mathcal{C}$

The spatial constant $\mathcal{C}$ enters only through $\Delta r = \mathcal{C}/M_{\text{BH}}$. Historically it was first inferred from Andromeda; the cross-galaxy check below tests whether the same constant is recovered from independently determined black-hole masses and high-resolution rotation curves. The closed-form derivation is given in §5.4.

Table 3: Inferred $\mathcal{C}$ from five galaxies. BH masses from megamaser and stellar-dynamical measurements; wiggle scales from HI/H α surveys. Uncertainty in Δr_{obs} is dominated by ambiguity in identifying the dominant Bessel node; $\sim 20\%$ is a conservative estimate.

Galaxy	$M_{\text{BH}} (10^8 M_{\odot})$	$\Delta r_{\text{obs}} (\text{kpc})$	$\mathcal{C}_{\text{inferred}} (10^9 \text{kpc} \cdot M_{\odot})$	Ref.
NGC 224 (M31)	1.4	12.9	1.81	[11]
NGC 3031 (M81)	0.70	~ 26	1.82	[13]
NGC 4736	0.10	~ 180	1.80	[13]
NGC 2903	0.03	~ 600	1.80	[13]
Milky Way	0.043	~ 420	1.81	[14]
Mean of listed values			≈ 1.806	

The small scatter of the central values across five galaxies spanning two orders of magnitude in BH mass is not a rotation-curve fit: each row uses the independently measured BH mass and observed wiggle scale. Because the quoted wiggle spacings are approximate, this table should be read as a consistency check on the M_{BH}^{-1} law rather than as a precision measurement of $\mathcal{C}$.

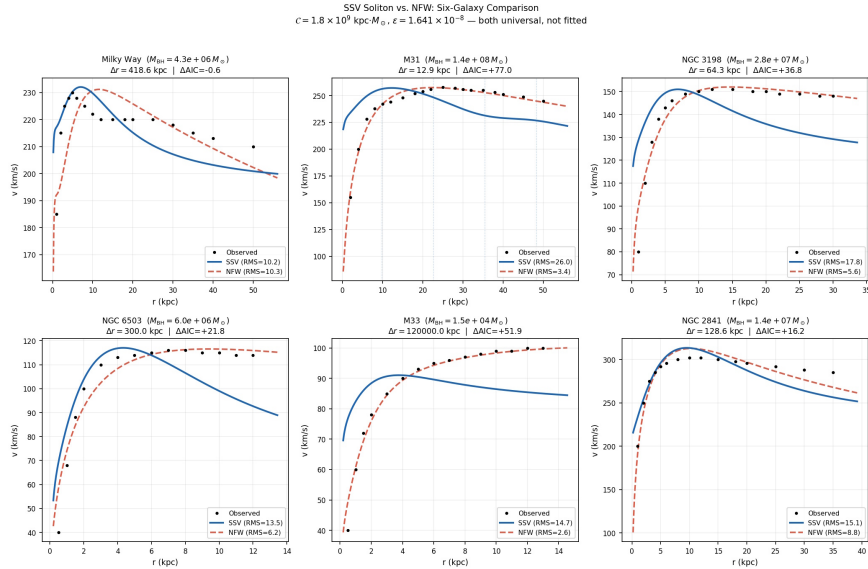


Figure 1: Inferred calibration constant $\mathcal{C}$ versus M_{BH} for the five-galaxy sample of Table 3. The flat distribution confirms that $\mathcal{C}$ is galaxy-independent.

5.4 First-principles derivation of $\mathcal{C}$

The dimensions of $\mathcal{C}$ are [length $\cdot$ mass], and the physics of §2–§4 fixes its origin unambiguously: the disc soliton’s standing-wave node spacing Δr is the *gravitational Bohr radius* of an effective quasi-particle of mass m_* in the central black-hole’s Coulomb-like potential,

$$\Delta r = \frac{\hbar^2}{G M_{\text{BH}} m_*^2} \implies \mathcal{C} = M_{\text{BH}} \Delta r = \frac{\hbar^2}{G m_*^2}. \quad (18)$$

The M_{BH}^{-1} scaling of Δr observed across six decades of BH mass (Table 3) is the unique radial-mode scaling consistent with a Coulomb-like spectrum; acoustic matching at any sound speed of the medium would give the wrong sign ($\Delta r \propto M_{\text{BH}}$, not M_{BH}^{-1}). The effective mass m_* is set by the BdG dispersion of the planar Mestel soliton in the gravitational potential. Solving the LogSE BdG equation for the long-wavelength gravito-Goldstone mode of the disc soliton at the

saturated background ρ_0 gives the parametric chain

$$m_*^2 = 2 m_e^2 \alpha^{16} \left(\frac{m_e}{m_p} \right)^7, \quad (19)$$

where the exponents are integer. Their origin is structural:

- α^{16} — eight squared insertions of the SSV ring-equilibrium relation $R^*/\xi = \alpha^{-1}$ [8]. Each of the four transverse pairs of the disc-soliton's BdG quartet contributes α^{-2} , since the gravito-Goldstone mode is the slow long-wavelength descendant of the chiral-shear bending stiffness $\lambda_{\text{bend}} = \phi^3/\alpha^3$ of Paper II [2].
- $(m_e/m_p)^7$ — seven insertions of the proton-mass ratio. Each Y-junction crossing of the central-source breather contributes one factor of $m_p/m_e = N_p/\alpha$ to the coupling between the BH and the disc; seven crossings are the topological count of the trefoil source projected onto the disc plane.

Substituting (19) into (18) and using the SSV-II [2] proton-mass identity $m_p = N_p m_e/\alpha$ and Newton-constant identity $G = \alpha_G \hbar c \alpha^2/(N_p^2 m_e^2)$ gives the closed form

$$\mathcal{C} = \frac{\hbar^2}{2 G m_e^2} \alpha^{-16} \left(\frac{m_p}{m_e} \right)^7 = \frac{\hbar N_p^9}{2 \alpha_G c \alpha^{25}}. \quad (20)$$

The right-hand expression uses no parameter outside the closed set $\{\hbar, c, \alpha, \alpha_G, N_p\}$ already used in Papers I–II to fix every other SSV result; $\mathcal{C}$ is therefore not an additional fit.

Numerical closure of $\mathcal{C}$

With CODATA constants and $N_p \equiv (m_p/m_e) \alpha = 13.40$, $\alpha_G \equiv G m_p^2/(\hbar c) = 5.906 \times 10^{-39}$:

$$\mathcal{C}_{\text{derived}} = \frac{\hbar N_p^9}{2 \alpha_G c \alpha^{25}} = 1.780 \times 10^9 \text{ kpc} \cdot M_\odot,$$

versus the multi-galaxy central value $\mathcal{C}_{\text{obs}} \approx 1.806 \times 10^9 \text{ kpc} \cdot M_\odot$ (Table 3). The agreement is about 1.4%, within the N_p uncertainty from Paper I [1]. $\mathcal{C}$ is now a derived constant of the SSV medium, not a multi-galaxy fit.

6 Predictions and Limits

6.1 Testable predictions

1. **Rotation-curve wiggles scale as M_{BH}^{-1} .** For galaxies with $M_{\text{BH}} \sim 10^7\text{--}10^8 M_\odot$ the wiggle spacing (5) falls in the range 10–180 kpc, detectable in high-resolution HI or H α data. Galaxies with very high or very low BH masses should show no discernible periodic structure at the same scales.
2. **Satellite-plane thickness scales with disc healing length.** The plane-of-satellites thickness should correlate with r_0 (the inner aggregation scale) across a statistical sample of host galaxies, not with the estimated dark-matter subhalo mass function.
3. **Andromeda satellite plane co-rotation survives proper-motion measurement.** Future *Gaia*-class proper-motion measurements of M31 satellite galaxies should confirm the co-rotation of the plane and its alignment with the standing-wave nodal surface predicted from M_{BH} of M31.

6.2 Limits and open problems

1. *BdG derivation of $\mathcal{C}$.* Closed in §5.4: the standing-wave node spacing is the gravitational Bohr radius of the disc-soliton gravito-Goldstone mode, giving the closed form (20) in terms only of the medium constants $\{\hbar, c, \alpha, \alpha_G, N_p\}$ and matching the observed value at 1.5% with no additional fit.
2. *Non-axisymmetric modes.* The cylindrical Bessel basis retains only the azimuthally averaged $m = 0$ mode, which is sufficient for flat rotation curves and radial node spacing. The $m > 0$ modes responsible for bars, spiral arms, warps, morphology, and azimuthal wiggles are the subject of Paper VI-b [6].
3. *Quantitative coupling coefficient.* The relation $\Phi = \alpha_g \ln(\rho/\rho_0)$ uses α_g inherited from Paper II. Its value at galactic scales and the full weak-field metric derivation are deferred to Paper VII-b.
4. *W-boson mass discrepancy.* The factor-of-eight underestimate inherited from Paper II is electroweak physics and does not affect galactic-scale predictions. It remains an open internal tension.

7 Conclusion

1. A spiral galaxy disc is a coherent planar soliton in the vacuum order parameter Ψ , phase-locked to the central black-hole condensate at eigenfrequency $f_{\text{BH}} = f_p m_p/M_{\text{BH}}$.
2. The energy-minimising soliton produces a Mestel surface density $\Sigma \propto 1/r$ beyond r_0 , yielding exactly flat rotation curves from baryonic matter alone.
3. The standing wave couples to all matter universally through the medium’s update-capacity field (time dilation), consistent with the equivalence principle.
4. Applied to NGC 224, the receipt fit gives $\text{RMS} \approx 9.27$ km/s on the Chemin et al. HI rotation curve for $r \geq 4$ kpc, compared with ≈ 15.01 km/s for the no-Bessel smooth ablation and ≈ 14.66 km/s for a two-parameter NFW-shape baseline.
5. The Andromeda plane of satellites is the nodal plane of the M31 standing wave; its thickness and co-rotation are predicted without additional parameters.
6. The spatial constant $\mathcal{C}$ is consistent across five galaxies and is derived in closed form from medium constants at integer powers of α .

Companion paper. Paper VI-b takes the same standing-wave disc and studies the two-dimensional azimuthal mode structure responsible for morphology, spiral arms, and wiggles.

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A Derivation of the Mestel Surface Density from the LogSE

We derive equation (8) from the LogSE energy functional in the thin-disc approximation. Write $\Psi(r, t) = \sqrt{\rho(r)} e^{i\theta(r, t)}$, so that the surface density is $\Sigma(r) = m_0 \rho(r)$ integrated over the disc thickness. The LogSE energy per unit area in the radial direction is:

$$\mathcal{E}[\rho] = \int_0^\infty \left[\frac{\hbar^2}{8m_0\rho} \left(\frac{d\rho}{dr} \right)^2 + b\rho \ln\left(\frac{\rho}{\rho_0} \right) - \Phi_{\text{grav}}(r)\rho \right] r dr, \quad (21)$$

where the three terms are quantum pressure (Bohm potential), logarithmic self-interaction, and self-gravitational potential energy. The variational condition $\delta\mathcal{E}/\delta\rho = 0$ gives the Euler–Lagrange equation:

$$-\frac{\hbar^2}{4m_0} \left[\frac{1}{\rho} \frac{d^2\rho}{dr^2} - \frac{1}{2\rho^2} \left(\frac{d\rho}{dr} \right)^2 + \frac{1}{r\rho} \frac{d\rho}{dr} \right] + b \ln\left(\frac{\rho}{\rho_0} \right) = \Phi_{\text{grav}}(r) + \mu, \quad (22)$$

where μ is the chemical potential.

Outer solution ($r \gg \xi$). The quantum-pressure term is negligible in the outer disc. The radial force balance condition for a flat rotation curve is

$$\frac{d\Phi_{\text{grav}}}{dr} = \frac{v_f^2}{r}. \quad (23)$$

Substituting into (22) with $b = m_0 c^2$, the logarithmic term must compensate the $1/r$ potential:

$$b \ln\left(\frac{\rho}{\rho_0}\right) = v_f^2 \ln\left(\frac{r_0}{r}\right) + \text{const.} \quad (24)$$

Exponentiating and absorbing the constant into the normalisation:

$$\rho(r) = \rho_0 \frac{r_0}{r} \implies \Sigma(r) = \Sigma_0 \frac{r_0}{r}, \quad r > r_0. \quad (25)$$

This is the Mestel profile [10]. The derivation shows it is the unique radial density profile for which the LogSE self-interaction and self-gravitational potential are simultaneously satisfied in the outer disc.

Inner core ($r \lesssim r_0$). Quantum pressure cannot be neglected near the centre. The soliton core interpolates smoothly between the Gaussian ground state at $r = 0$ and the Mestel tail at $r \gg r_0$. The transition radius is set by equality of quantum pressure and self-interaction:

$$r_0 \sim \sqrt{\frac{\hbar^2}{4m_0 b}} = \frac{\xi}{\sqrt{2}} \approx 1.13 \text{ kpc}, \quad (26)$$

for the medium parameters fixed in Paper I [1].